Chapter 1

Series of Functions

Using the tools developed in the previous chapter, we may now discuss the concept of series of functions, for example Taylor series are series of polynomials x^n. More importantly, we may now consider when such series converge uniformly to a limit function, defined in terms of the series. From our discussion of Taylor series, we introduced analytic functions which converge *pointwise* to a limit function $\sum_{n=0}^{\infty} a_n x^n$ on some interval of $\mathbb{R}$. By the end of this chapter, we will be able to analyse when such Taylor series converge uniformly.

If (φ_n) is a sequence of functions on $[a, b]$ then we define the *partial sums*

$$f_n(x) \coloneqq \sum_{k=1}^{n} \varphi_k(x), \quad x \in [a, b].$$

Definition 1.1 (Uniform Convergence of Series)**.** We say that the series $\sum_{n=1}^{\infty} \varphi_n$ *converges uniformly* on $[a, b]$ to a function f if the sequence (f_n) converges uniformly on $[a, b]$. In that case, we write

$$f(x) = \sum_{n=1}^{\infty} \varphi_n(x),$$

meaning that $f_n \to f$ uniformly.

The definition above is just the usual definition of uniform convergence, applied to the sequence of partial sums (f_n). In particular, we can apply the theorems from last section. In particular, we have the following three corollaries from results established in the previous chapter.

Corollary 1.2. *Let (φ_n) be a sequence of continuous functions on $[a, b]$. If $f_n \coloneqq \sum_{k=1}^{n} \varphi_k \rightrightarrows f$ then f is continuous.*

Proof. Since each φ_n is continuous, each f_n is continuous by the Algebra of Continuous Functions (Theorem **??**). Since $f_n \to f$ uniformly, Theorem **??** implies that $f = \sum_{n=1}^{\infty} \varphi_n$ is continuous on $[a, b]$. □

Corollary 1.3. *Let (φ_n) be a sequence of integrable functions on $[a, b]$. If $f_n := \sum_{k=1}^{n} \varphi_k \rightrightarrows f$ then f is integrable and*

$$\int_a^b f(x)\,\mathrm{d}x = \int_a^b \sum_{n=1}^{\infty} \varphi_n(x)\,\mathrm{d}x = \sum_{n=1}^{\infty} \int_a^b \varphi_n(x)\,\mathrm{d}x.$$

Proof. Since each φ_n is integrable, by linearity of the integral each f_n is integrable and, for each $n \in \mathbb{N}$,

$$\int_a^b f_n(x)\,\mathrm{d}x = \sum_{k=1}^{n} \int_a^b \varphi_k(x)\,\mathrm{d}x.$$

Applying Theorem **??** by uniform convergence of f_n to f gives

$$\int_a^b f(x)\,\mathrm{d}x = \lim_{n\to\infty} \int_a^b f_n(x)\,\mathrm{d}x = \lim_{n\to\infty} \sum_{k=1}^{n} \int_a^b \varphi_k(x)\,\mathrm{d}x = \sum_{n=1}^{\infty} \int_a^b \varphi_n(x)\,\mathrm{d}x.$$

□

Corollary 1.4. *Let (φ_n) be a sequence of functions in $C^1([a, b])$, and suppose:*

1. *For some $x_0 \in [a, b]$, the series $\sum_{n=1}^{\infty} \varphi_n(x_0)$ converges;*
2. *The series of derivatives $\sum_{n=1}^{\infty} \varphi_n'$ converges uniformly on $[a, b]$.*

Then the partial sums $f_n := \sum_{k=1}^{n} \varphi_k$ converge uniformly on $[a, b]$ to a function f, which is differentiable on (a, b) and satisfies

$$f'(x) = \sum_{n=1}^{\infty} \varphi_n'(x) \quad \forall x \in (a, b),$$

where the series on the right is understood as the uniform limit of its partial sums.

1.1 Weierstraß M-Test

In practice, it is often inconvenient to verify uniform convergence directly from the definition, so we would like a sufficient condition that implies uniform convergence of series that is easier to work with. The Weierstraß M-test provides a simple sufficient criterion: if each term (or each derivative term) can be dominated by a numerical series that converges, then the relevant uniform convergence follows automatically.

Definition 1.5 (Uniformly Cauchy sequence)**.** Let $f_n : D \to \mathbb{R}$ be a sequence of functions. We say that the sequence of functions (f_n) is *uniformly Cauchy* if

$$\forall \varepsilon > 0,\ \exists N \in \mathbb{N},\ \forall m, n \geq N,\ \|f_m - f_n\|_\infty < \varepsilon.$$

Lemma 1.6. *Let $f_n : D \to \mathbb{R}$ be a sequence of functions. Then (f_n) converges uniformly if and only if (f_n) is uniformly Cauchy.*

Proof. ($\Rightarrow$) Suppose $f_n \to f$ uniformly. Given $\varepsilon > 0$, by definition there exists $N \in \mathbb{N}$ such that for all $n \geq N$, $\|f_n - f\|_\infty < \frac{\varepsilon}{2}$. Therefore, by the triangle inequality for the supremum norm we have, for $m, n \geq N$,

$$\|f_m - f_n\|_\infty \leq \|f_m - f\|_\infty + \|f_n - f\|_\infty < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

($\Leftarrow$) Conversely, suppose (f_n) is uniformly Cauchy. Then in particular, the sequence $(f_n(x))$ is Cauchy for all $x \in D$ and since $\mathbb{R}$ is complete, $f_n(x)$ therefore converges for all $x \in D$. Construct $f : D \to \mathbb{R}$ by letting $f(x) = \lim_{n\to\infty} f(x)$ for all $x \in D$. We now want to show that $f_n \to f$ uniformly. Given $\varepsilon > 0$, let $N \in \mathbb{N}$ be such that for all $m, n \geq N$ we have $\|f_m - f_n\|_\infty < \frac{\varepsilon}{2}$. In particular, fixing $x \in D$ and $n \in \mathbb{N}$ and letting $m \to \infty$ yields

$$|f_n(x) - f(x)| \leq \frac{\varepsilon}{2},$$

since $f_m(x) \to f(x)$ as $m \to \infty$. Thus for all $x \in D$, $|f_n(x) - f(x)| \leq \frac{\varepsilon}{2} < \varepsilon$, so $f_n \to f$ uniformly. □

This is an important result. If we are given a concrete sequence of functions, then the usual way to show it converges is to compute the pointwise limit and then prove that the convergence is uniform. However, if we are dealing with sequences of functions in general, this is less likely to work. Instead, it is often much easier to show that a sequence of functions is uniformly convergent by showing it is uniformly Cauchy.

Theorem 1.7 (Weierstraß M-test)**.** *Let (φ_n) be a sequence of functions on $[a, b]$ and define $f_n := \sum_{k=1}^{n} \varphi_k$ as the partial sums. Suppose there exists numbers $M_n \geq 0$ such that $\|\varphi_n\|_\infty \leq M_n$, and $\sum_{n=1}^{\infty} M_n$ converges. Then $f_n \rightrightarrows f$ on $[a, b]$.*

Proof. First we must show that there exists a limit function, f, since we do not yet have a candidate limit. We do this by using the Cauchy criterion for sequences of functions. Given natural numbers $m < n$ and $x \in [a, b]$, we have, by the triangle inequality and hypothesis,

$$|f_m(x) - f_n(x)| = \left| \sum_{k=n+1}^{m} \varphi_n(x) \right| \leq \sum_{k=n+1}^{m} |\varphi_k(x)| \leq \sum_{k=n+1}^{m} M_k \leq \sum_{k=n+1}^{\infty} M_k.$$

Since $\sum_{n=1}^{\infty} M_n$ converges, we know that the tail $\sum_{k=n}^{\infty} M_k \to 0$ as $n \to \infty$, so given $\varepsilon > 0$, $N \in \mathbb{N}$ can be chosen sufficiently large such that for all $n \geq N$, $\sum_{k=n+1}^{\infty} M_k < \varepsilon$. Therefore, (f_n) is uniformly Cauchy so by Lemma 1.6, $f_n \to f$ uniformly on $[a, b]$. □

Example 1.8. Let $\varphi_n : [0, 1] \to \mathbb{R}$ be defined by $\varphi_n(x) = \frac{\cos(nx)}{n^2}$. Then $\|\varphi_n\|_\infty \leq \frac{1}{n^2}$ for each $n \in \mathbb{N}$, and the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges by the p-series test since $p = 2 > 1$. Therefore, the partial sums $f_n := \sum_{k=1}^{n} \varphi_k$ converge uniformly to $f := \sum_{n=1}^{\infty} \varphi_n$ on $[0, 1]$. The limit function $f : [0, 1] \to \mathbb{R}$ defined by $f(x) = \sum_{n=1}^{\infty} \frac{\cos(nx)}{n^2}$ is called a *Fourier series.*

By Corollary 1.4, this Fourier series can be differentiated term-by-term to yield a new Fourier series (since cos differentiates to $-\sin$), $g : [0, 1] \to \mathbb{R}$ defined by

$$g'(x) = f(x) = -\sum_{n=1}^{\infty} \frac{\sin(nx)}{n}.$$

Note: Fourier series and their computation is explored in Calculus and Applications (MATH40004).

1.2 Uniform Convergence of Power Series

A particularly interesting kind of series is a power series, and we have already met these in Chapter 7, and proved some results about them. However, as mentioned in the introduction to this chapter, our results were pointwise results, discussing how the series behaves at a particular point $x \in \mathbb{R}$. Here we will quickly look into how the power series behaves as a function of x. In particular, we want to know whether it converges uniformly.

Theorem 1.9. *Let $\sum_{n=0}^{\infty} c_n(x-a)^n$ be a (real) power series. Then there exists a unique number $R \in [0, +\infty]$ (called the radius of convergence) such that*

1. *If $|x-a| < R$, then $\sum c_n(x-a)^n$ converges absolutely;*
2. *If $|x-a| > R$, then $\sum c_n(x-a)^n$ diverges;*
3. *If $R > 0$ and $0 < r < R$ then $\sum c_n(x-a)^n$ converges uniformly on $[a-r, a+r]$.*

Proof. Items (1) and (2) were proven in Chapter 2. For (3), note that from (1), taking $x = a-r$ shows that $\sum |c_n|r^n$ is convergent. But if $x \in [a-r, a+r]$ then $|x-a| \leq r$ so

$$|c_n(x-a)^n| \leq |c_n|r^n,$$

so the result follows by the Weierstraß M-test (Theorem 1.7) with $M_n = |c_n|r^n$ which is independent of x. $\square$

Note that these results hold for complex power series as well. Also, uniform convergence need not hold on the entire interval of convergence, as the following example will demonstrate.

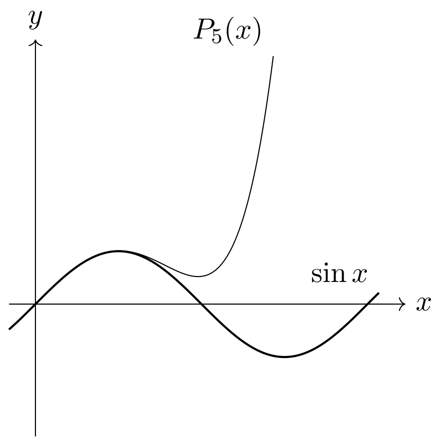
Example 1.10. Consider the power series $\sum x^n$. The interval of convergence of this power series is $(-1, 1)$, however it does not converge *uniformly* on $(-1, 1)$. For $m < n$, the tail

$$\sum_{k=m}^{n} x^k = x^n \sum_{k=0}^{n-m} x^k \geq \frac{x^m}{1-x},$$

for $x \in (-1, 1)$. Since uniform convergence for series of functions relies on this tail going to zero uniformly in x, but the above expression on the right hand side can be made arbitrarily small by choosing x close to 1, we do not have uniform convergence on $(-1, 1)$.

Remark. The idea for why power series do not necessarily converge on their entire *open* interval of convergence is the same as the idea for why uniform continuity may only work on a compact interval and not the full open interval, for example the function $x \mapsto \frac{1}{x}$ on $(0, 1)$ is not uniformly continuous, but is uniformly continuous on $[r, 1)$ for any $0 < r < 1$. It is due to the fact that, with open intervals, there are limit points which are not in the interval itself, where points x in the interval can "escape" to. If these points are where a function has a singularity (for example $\frac{1}{1-x}$ in the above example, near $x = 1$), then we cannot control *uniformly* in x in the open interval, because by virtue of the interval being *open*, one can always choose a point closer to the singularity, breaking the notion *uniform* convergence (we are required to find some $N \in \mathbb{N}$, that works for all x in the interval, whereas pointwise convergence relies on choosing an $N \in \mathbb{N}$ that may depend on ε and x).

Another way to see this, specifically for Taylor series, is that polynomials have tails that blow up to $\pm\infty$ eventually, so if this is the Taylor series of, say, a bounded function on $\mathbb{R}$, we cannot possibly expect the Taylor series to converge uniformly on $\mathbb{R}$, because for each finite $N \in \mathbb{N}$ we choose, the Taylor polynomial approximation will blow up eventually for some large x, but the function is bounded, so while pointwise convergence holds here, uniform convergence does not, because no matter which $N \in \mathbb{N}$ we choose, there is always a large x such that the Taylor polynomial has blown up and has "stripped away" from the target function. This is illustrated below with the function $x \mapsto \sin x$ on $\mathbb{R}$.



1.3 Beyond the Weierstraß M-Test

One should also remember that the M-test is only a sufficient condition; uniform convergence may still hold if no summable supremum bound exists.

For instance, two different mechanisms by which we can construct sequences of functions which converge uniformly are:

- *Geometric structure*: the functions in the sum have (almost) disjoint supports, so that at each point only few functions are "active";
- *Cancellation*: (e.g. alternating signs) where large terms cancel in the partial sums.

The next two examples illustrate these two phenomena.

Example 1.11 (Disjoint supports: uniform convergence without a summable bound on $|\phi_n'|$.)**.** We will work on the compact interval $[0, 1]$. Define

$\operatorname{dist}(t,\{1,2\}) := \min\{|t-1|,|t-2|\}$ to be the shortest distance from $t \in [1,2]$ to the points in the set $\{1,2\}$, and define the continuous "tent" function $\tau : \mathbb{R} \to \mathbb{R}$ by

$$\tau(t) := \begin{cases} \operatorname{dist}(t,\{1,2\}), & t \in [1,2], \\ 0, & t \notin [1,2]. \end{cases}$$

Therefore, the *support* of τ (that is, the set of values of t for which $\tau(t) \neq 0$) is contained in $[1,2]$, and the supremum of τ is $\frac{1}{2}$.

Now we will construct our sequence of functions. For $n \geq 1$, define

$$\psi_n(x) := \frac{2}{n}\tau(2^{n+1}x), \quad x \in [0,1],$$

and set

$$\phi_n(x) := \int_0^x \psi_n(t)\,\mathrm{d}t,$$

so that in particular $\phi_n' = \psi_n$ by the Fundamental Theorem of Calculus (Theorem **??**).

Now, notice that since τ vanishes outside of the interval $[1,2]$, the support of each ψ_n is the set of values of x for which

$$1 < 2^{n+1}x < 2,$$

i.e. $(2^{-(n+1)}, 2^{-n})$ is the support of ψ_n. These intervals are pairwise disjoint.

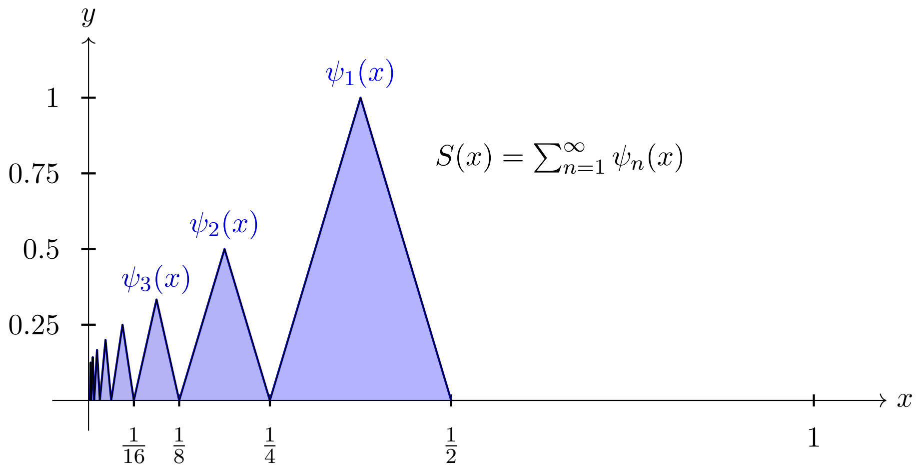
Uniform convergence of the derivative series. Because $\sup|\tau| = \frac{1}{2}$, we have $\sup|\psi_n| = \frac{1}{n}$ by construction. Fixing $N \geq 1$, for any $x \in [0,1]$ we see that at most one term $\psi_n(x)$ with $n \geq N+1$ is non-zero, so for all $m > N$ we have

$$\left|\sum_{n=N+1}^{m} \phi_n'(x)\right| = \left|\sum_{n=N+1}^{m} \psi_n(x)\right| \leq \sup_{n \geq N+1} |\psi_n| = \frac{1}{N+1}.$$

Thus the derivative series $\sum \phi_n'$ is uniformly Cauchy, so it converges uniformly on $[0,1]$.

The uniform limit function is visualised below; the idea behind the construction is that each $x \in [0,1]$ belongs to precisely one interval of the form $(2^{-(n+1)}, 2^{-n})$, on which only the function ψ_n is supported, so the infinite sum reduces to $\psi_n(x)$. The process of uniform convergence, here, acts as a "glue" stitching infinitely many localized functions ψ_n together across the domain.

The geometric separation of the functions guarantees that uniform convergence relies exclusively on the spatial disjointness of the functions; because at most one term in the sum is non-zero for any x, the absolute value of the tail of the series reduces to the absolute value of a single active term. The looseness of the M-test arises because replacing $|\psi_n(x)|$ with $\sup|\psi_n|$ discards the spatial dependence of each function. The final bound represents the sum of the global suprema regardless of where those values are attained. In this case we were able to construct a uniformly convergent function where this sup-norm summation is $\sum \frac{1}{n}$, which famously diverges.



Uniform convergence of the original series. Each ϕ_n is increasing and non-negative, so $\sup|\phi_n| = \phi(1)$. Moreover,

$$\sup|\phi_n| = \int_0^1 \psi_n(t)\,\mathrm{d}t = \frac{2}{n}\int_0^1 \tau(2^{n+1}t)\,\mathrm{d}t,$$

and with the substitution $u = 2^{n+1}t$ we obtain

$$\sup|\phi_n| = \frac{2}{n}2^{-(n+1)}\int_1^2 \mathrm{dist}(u,\{1,2\})\,\mathrm{d}u.$$

Now, by the formulae for areas of triangles, the value of the integral is $\frac{1}{4}$ (tent of base 1, height $\frac{1}{2}$), so $\sup|\phi_n| = \frac{2^{-(n+2)}}{n}$. Since $\sum \frac{2^{-(n+2)}}{n}$ converges by the ratio test, the Weierstraß M-test (Theorem 1.7) implies that $\sum \phi_n$ converges uniformly on $[0,1]$.

Example 1.12 (Cancellation: uniform convergence although $\sum \sup|\phi_n|$ and $\sum \sup|\phi_n'|$ diverge)**.** Again, we work on $[0,1]$. For $n \geq 1$, define

$$\phi_n(x) := (-1)^{n-1}\frac{x}{n+x}, \quad x \in [0,1].$$

Then $\phi_n \in C^1([0,1])$ and

$$\phi_n'(x) = (-1)^{n-1}\frac{n}{(n+x)^2}.$$

Uniform convergence of $\sum \phi_n$ (alternating series mechanism). Let $u_n(x) := \frac{x}{n+x}$. For each fixed $x \in [0,1]$, the sequence $u_n(x)$ is decreasing in n and $u_n(x) \to 0$ as $n \to \infty$. Moreover,

$$0 \leq u_n(x) \leq \sup_{t\in[0,1]} u_n(t) = \frac{1}{n+1},$$

so using the alternating series estimate gives, for every $N \geq 1$ and every $x \in [0,1]$,

$$\left|\sum_{n=N+1}^{\infty} \phi_n(x)\right| \leq u_{N+1}(x) \leq \frac{1}{N+2}.$$

Hence, $\sum \phi_n$ converges uniformly on $[0,1]$. *Uniform convergence of* $\sum \phi_n'$. Set $v_n(x) := \frac{n}{(n+x)^2}$. For each fixed $x \in [0,1]$, the function $s \mapsto \frac{s}{(s+x)^2}$ is decreasing for $s \geq x$; in particular, $v_n(x)$ is decreasing in n for all $n \geq 2$. Also, $0 \leq v_n(x) \leq v_n(0) = \frac{1}{n}$, so $v_n(x) \to 0$ uniformly. Applying the alternating series estimate again to $\sum_{n=2}^{\infty}(-1)^{n-1} v_n(x)$ yields, for every $N \geq 2$ and $x \in [0,1]$,

$$\left|\sum_{n=N+1}^{\infty} \phi_n'(x)\right| \leq v_{N+1}(x) \leq \frac{1}{N+1},$$

so $\sum \phi_n'$ converges uniformly on $[0,1]$.

Failure of the M-test. We have

$$\sup|\phi_n| = \sup_{x\in[0,1]} \frac{x}{n+x} = \frac{1}{n+1} \quad \text{and} \quad \sup|\phi_n'| = \sup_{x\in[0,1]} \frac{n}{(n+x)^2} = \frac{1}{n},$$

so both $\sum \sup|\phi_n|$ and $\sum \sup|\phi_n'|$ diverge, so the Weierstraß M-test again does not apply, despite uniform convergence of both series.

Finally, note also that Since $\sum \phi_n(0) = 0$ and $\sum \phi_n'$ converges uniformly, Corollary 1.4 still applies: the sum $f := \sum \phi_n$ is differentiable on $(0,1)$ and $f' = \sum \phi_n'$ there.